

BIOTECHNOLOGY: DNA EXTRACTION



Discover the tiny world inside you!

WHAT IS BIOTECHNOLOGY?



Biotechnology means using living things to solve problems and make useful products!

Examples include making medicines to help sick people, improving crops for farmers, and cleaning up the environment. Scientists use tiny living things to create amazing solutions!



MEET DNA — THE BUILDING BLOCKS OF LIFE

DNA carries the instructions for how all living things grow and function. Think of it like a recipe book inside every cell that tells your body how to build and run itself!

WHY EXTRACT DNA?

Scientists extract DNA for many important reasons:

01

To study genes and traits that make us unique

02

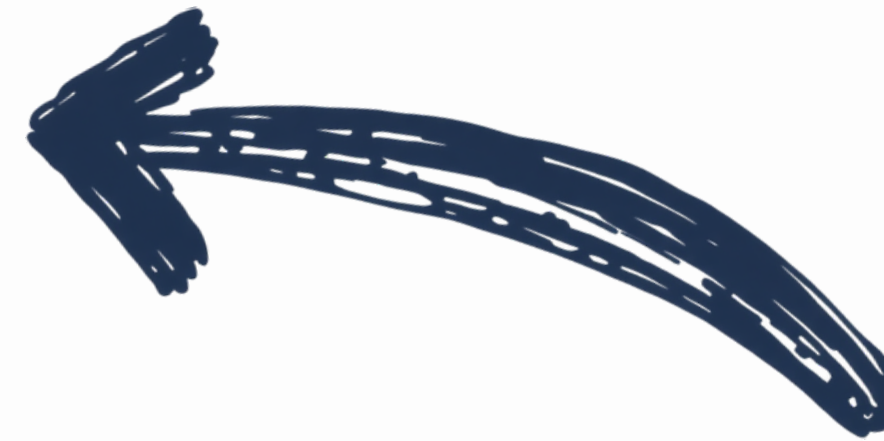
To solve crimes using forensic science evidence

03

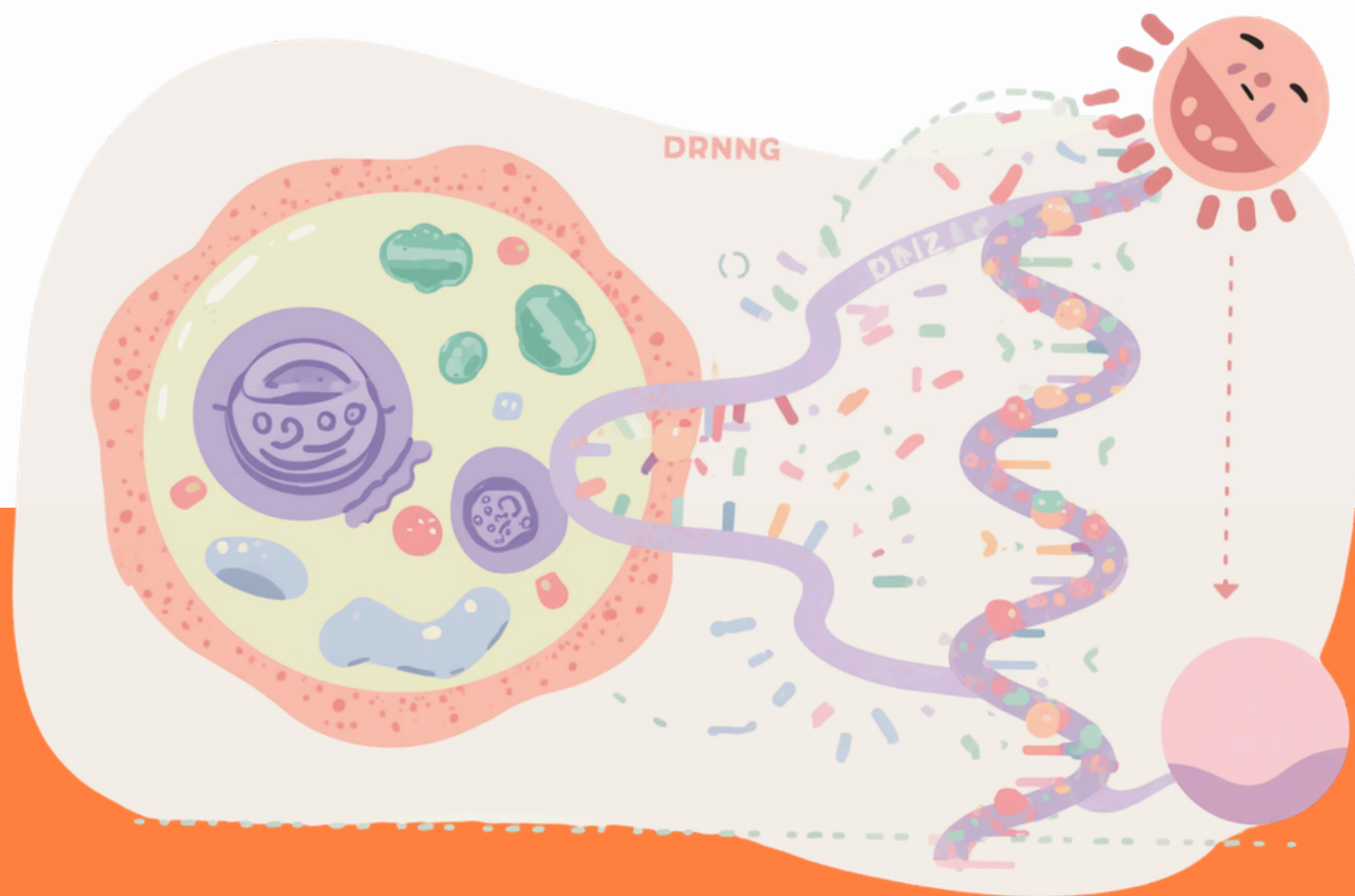
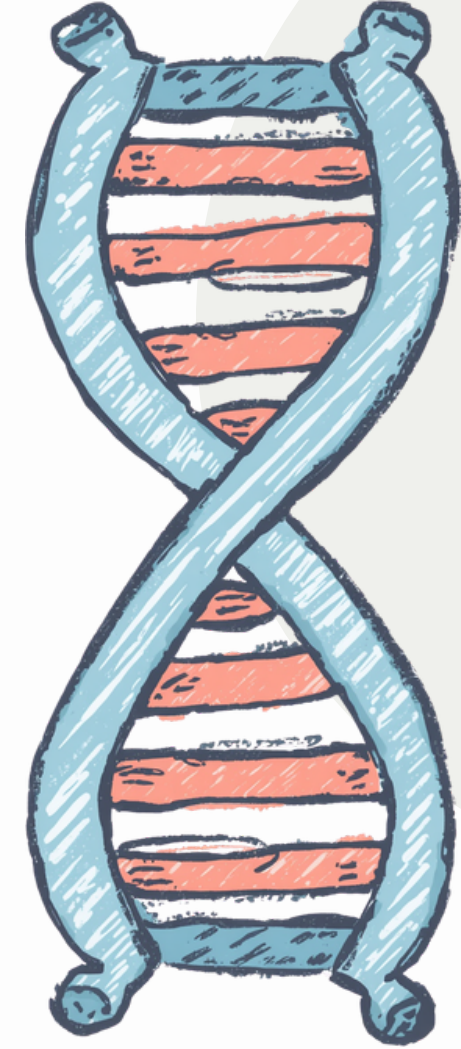
To improve crops and develop new medicines for people

04

To learn about living things and how they work



WHAT IS DNA EXTRACTION?



DNA extraction means separating DNA from the rest of the cell parts.

This process helps scientists see and study DNA clearly. By removing DNA from cells, we can examine the tiny instructions that make every living thing unique!

MATERIALS NEEDED FOR DNA EXTRACTION



01

Soap or detergent
to break open cell
walls

02

Salt to help clump
the DNA together

03

Water and fruit like
strawberries or
bananas





MATERIALS NEEDED FOR DNA EXTRACTION (PART 2)



04



Coffee filter or
cloth to strain
the mixture

05

Chilled rubbing
alcohol or ethanol

06

Clear container or
test tube to collect
DNA



STEP 1 — PREPARE THE FRUIT SAMPLE



Mash the fruit gently to break open the cells. This releases the DNA from inside! Use a fork or your hands to squish strawberries or bananas in a bowl until smooth.



STEP 2 — ADD SOAP SOLUTION

Mix soap, salt, and water together to create a special solution. This mixture helps dissolve the cell membranes and release the DNA from inside the fruit cells.

STEP 3 — FILTER THE MIXTURE



Pour the mixture through a coffee filter or cloth to remove the solid bits. This leaves you with a clear liquid that contains the DNA!



STEP 4 — ADD COLD ALCOHOL

Slowly pour cold alcohol on top of your filtered mixture. The alcohol makes the DNA come out of the liquid as white, stringy threads that you can see!

STEP 5 — OBSERVE THE DNA



White, stringy DNA will appear between the layers! Use a stick or glass rod to gently lift the DNA strands out. You can now see the tiny instructions that make living things work!

FUN FACTS ABOUT DNA

Amazing things you should
know about DNA:

01

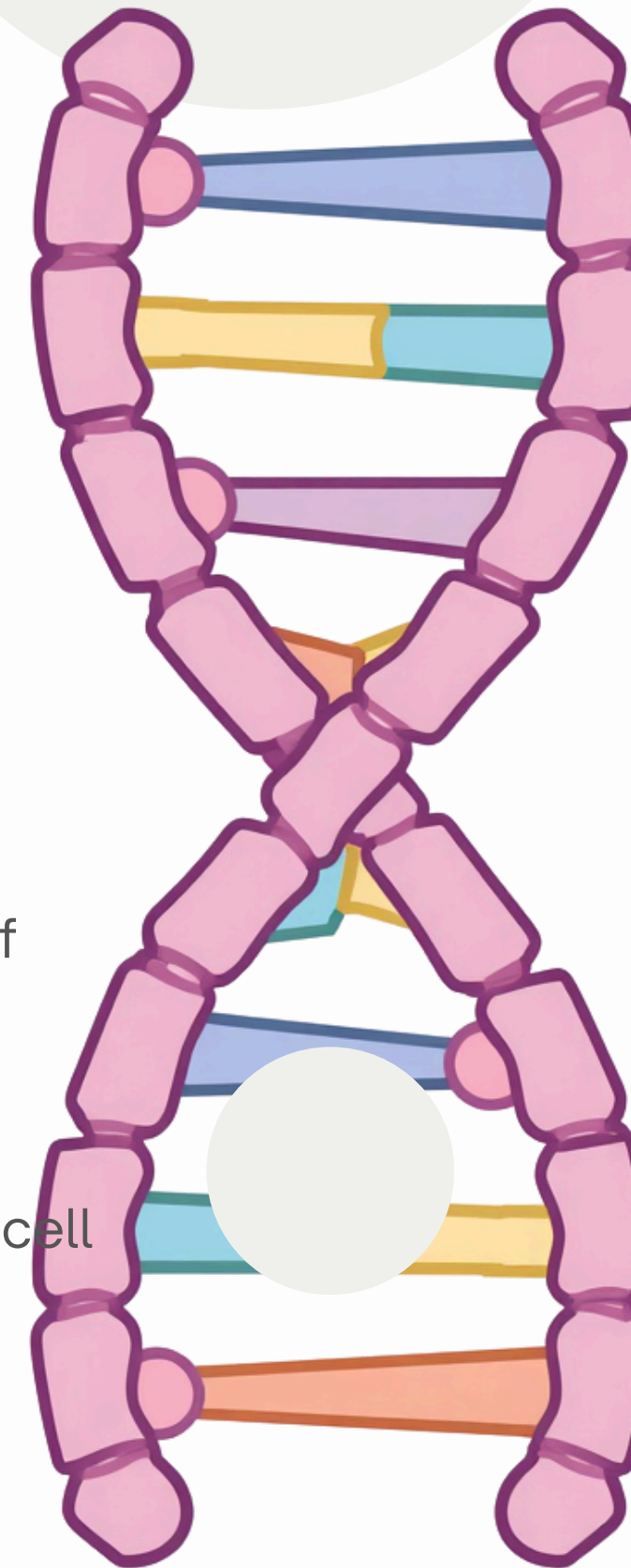
DNA is shaped like a
twisted ladder called a
double helix

02

Humans share about 99.9% of
DNA with each other

03

DNA is found in almost every cell
of your body



REAL-LIFE APPLICATIONS OF DNA EXTRACTION



DNA extraction helps scientists and experts solve important problems in many fields.

- Solving crimes through forensics
- Helping farmers grow better and stronger crops
- Developing new medicines to cure diseases
- Understanding genetic conditions

REMEMBER THESE SAFETY TIPS

Follow these important guidelines to stay safe during your DNA extraction experiment:



01

Always ask an adult before handling chemicals

03

Wash hands thoroughly after the experiment

02

Use clean equipment for accurate results

04

Work carefully and slowly throughout the process

SUMMARY — WHAT WE LEARNED TODAY

Key takeaways from our DNA extraction journey:



16

Biotechnology helps us understand and use living things

16

DNA is the instruction book inside every cell

16

DNA extraction uses simple materials like soap and alcohol

THANK YOU!!



Thank you for exploring DNA with us! Try the experiment safely at home or school, and keep asking questions!

